

Psycho-social characteristics of the upper echelons

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Abstract

According to Upper Echelons Theory, organizational outcomes are reflections of top executives' values and cognitive bases (Hambrick & Mason, 1984). Among the executive psychosocial characteristics that have attracted substantial scholarly attention, narcissism, hubris, and overconfidence are frequently conflated in the literature as they share a common cognitive root in self-enhancement bias, the systematic tendency to overestimate one's own

judgment and underestimate external risk. This conceptual conflation has obscured the distinct psychological mechanisms through which each construct shapes strategic behavior and firm performance. To address this theoretical gap, this review poses two core research questions: (1) How do CEO narcissism, hubris, and overconfidence resemble and differ from one another in terms of construct definition, psychological mechanism, and strategic behavior? (2) How do the differences among these three forms of bias shape the logic of governance intervention? To answer these questions, this review synthesizes 205 studies. This review develops a Same Root, Divergent Forms integrative framework. The divergent forms refer to three distinct psychological mechanisms through which this shared drive produces differentiated strategic behavior. Narcissistic CEOs pursue external admiration and attention. Hubristic CEOs systematically exclude information inconsistent with their existing beliefs. Overconfident CEOs overestimate the accuracy of their own judgment. These three pathways introduce bias at distinct stages of the decision-making chain, generating differentiated strategic choice and performance consequences. The framework further reveals that identical governance mechanisms produce different constraining effects across constructs, depending on whether they engage the specific cognitive fault point at which each construct operates.

Keywords: Upper Echelons Theory, CEO narcissism, CEO hubris, CEO overconfidence, integrative review

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1. Introduction

Within the management literature, upper echelons theory (UET) offers a foundational perspective on firm performance: CEO psycho-social characteristics shape organizational outcomes by filtering and influencing strategic decisions (Bromiley & Rau, 2016; Hambrick & Mason, 1984). This proposition established the theoretical standing of executive psychology within organizational research and generated a large body of empirical work on the relationships among CEO personality traits, cognitive biases, and strategic choices. As this research accumulated, the measurement of CEO psycho-social characteristics evolved alongside it: from early indirect operationalizations based on archival data (Chatterjee & Hambrick, 2007; Hayward & Hambrick, 1997) to direct survey scales (Dick et al., 2021; Li & Tang, 2010; Park et al., 2011; Wales et al., 2013), video-based ratings (Petrenko et al., 2016), and textual and linguistic analysis (Akstinaite et al., 2020; Craig & Amernic, 2018). A substantial body of evidence now demonstrates that CEO psycho-social characteristics produce differential effects on strategic behavior and firm performance (Saesen et al., 2024; Wales et al., 2013).

CEO dark personality characteristics have attracted substantial scholarly attention (Palmer et al., 2020; Peterson et al., 2012; Picone et al., 2014; Shirokova et al., 2023). Among the range of dark personality constructs, this review focuses on narcissism (the need for external admiration), hubris (contempt for disconfirming evidence), and overconfidence (overestimation of one's own judgment), because all three share a common cognitive root in self-enhancement bias: the systematic tendency of CEOs to overestimate their own capability and to underestimate external risk constraints (Chatterjee & Hambrick, 2007; Hayward & Hambrick, 1997; Malmendier & Tate, 2005). This shared root distinguishes them

fundamentally from the dark triad of personality, which comprises narcissism, Machiavellianism, and psychopathy (Paulhus & Williams, 2002). The core of Machiavellianism is strategic manipulation: the instrumental use of others as means to personal ends (Jones & Paulhus, 2014). The core of psychopathy is impulsive callousness, an absence of empathy, and disregard for moral norms (Hare, 1996). Both constructs are oriented toward interpersonal exploitation and emotional detachment rather than cognitive self-overestimation. Within the theoretical domain of CEO self-enhancement, narcissism, hubris, and overconfidence therefore constitute a conceptual cluster defined by a shared cognitive root but differentiated by distinct psychological mechanisms.

The existing empirical literature on CEO narcissism, hubris, and overconfidence has accumulated substantial evidence, yet it confronts two interrelated limitations. First, the conceptual boundaries among the three constructs have remained persistently unclear. Some studies treat hubris and overconfidence as equivalent constructs and use them interchangeably (Bollaert & Petit, 2010; Li & Tang, 2010), producing systematic confusion in the selection of measurement instruments and the interpretation of effects across studies. Second, the majority of research examines each construct in isolation, without cross-construct integration or comparison. It therefore remains unclear whether narcissism, hubris, and overconfidence generate different strategic effects and whether they require different governance responses. This gap prevents the development of targeted theoretical guidance for practice. These two limitations together define the core problem that this integrative review is designed to address.

To address these limitations, this review develops what I term a Same Root, Divergent Forms integrative framework. The shared root refers to the common cognitive origin: all three constructs are driven by self-enhancement bias. The divergent forms refer to the distinct

psychological mechanisms through which this shared drive is expressed: narcissistic CEOs seek external recognition and attention; hubristic CEOs exclude information inconsistent with their existing beliefs; overconfident CEOs overestimate the probability of success. Each pathway generates bias at a distinct stage of the information-processing chain: attention allocation, information intake, and probability calibration respectively. These divergent mechanisms produce differentiated patterns of strategic choice and performance consequence. This review addresses two research questions: (1) In what ways do CEO narcissism, hubris, and overconfidence resemble and differ from one another at the level of construct definition, psychological mechanism, and strategic behavior? (2) How do the differences among these three forms of bias shape the logic of governance intervention?

This review employs the integrative review method (Cronin & George, 2020) to conduct a systematic search of the Web of Science database, restricting the population to journals rated at AJG 3 or above and to studies at the CEO, TMT, and board levels. Initial database searches returned 635 records for narcissism, 644 for overconfidence, and 140 for hubris (N = 1419 in total). After screening, 205 studies were retained: 88 on narcissism, 86 on overconfidence, and 31 on hubris. These studies are systematically compared along four dimensions: construct definition, measurement approach, strategic behavior, and performance consequences.

This review advances existing theory along three dimensions. First, it develops the Same Root, Divergent Forms integrative framework. This framework is the first to incorporate all three constructs within a single analytical structure, simultaneously accounting for both their similarity and their divergence at the level of mechanism. Second, it provides a more fine-grained account of the cognitive pathways posited by upper echelons theory (Hambrick & Mason, 1984), advancing the level of analysis from a trait-to-outcome model to a

four-stage sequence running from trait to cognitive fault point to strategic decision to organizational outcome. Third, it establishes the construct specificity of effective governance: identical monitoring and incentive mechanisms produce fundamentally different effects across the three constructs. The review concludes with a discussion of limitations and directions for future research.

2. Methodology

This paper adopts an integrative review design with a redirection purpose (Cronin & George, 2020) to synthesize three previously siloed research streams (narcissism, hubris, and overconfidence) within the Upper Echelons Theory framework (Hambrick & Mason, 1984) as the primary organizing lens, drawing on agency theory (Jensen & Meckling, 1976), stakeholder theory (Freeman, 1984), and the attention-based view (Ocasio, 1997) where relevant to specific strategic domains. This review is not a summary of three separate literatures. It asks what these constructs share at the cognitive level, how their mechanisms come apart, and what those differences imply for strategic behavior and governance.

Search Strategy. Systematic literature searches were conducted in Web of Science (Core Collection), the primary database for management and organizational behavior research. Searches were executed separately for each of the three constructs: (1) narcissism or narcissistic (2) overconfidence (3) hubris. At the point of retrieval, three database filters were applied to all searches: (1) document type restricted to Article, excluding editorials, book reviews, conference proceedings, and other non-peer-reviewed formats; (2) research area restricted to Business and Management; and (3) language restricted to English. These searches returned 635 records for narcissism, 644 for overconfidence, and 140 for hubris (N = 1419 in total).

Screening Criteria. Then we applied four eligibility criteria (Table 1) in a fixed sequence designed to maximize screening efficiency. We applied C1 first because journal quality can be verified instantly against the AJG 2024 list, enabling bulk elimination before any reading is required. C2 through C4 required progressively deeper engagement with the text and were therefore applied at the title/abstract stage; ambiguous cases were advanced to full-text review. We excluded a paper as soon as any single criterion was unambiguously violated.

Table 1 Eligibility Criteria

Code	Dimension	Inclusion	Exclusion
C1	Journal Quality	AJG 2024 rating of 3, 4, or 4*	AJG rating of 1-2
C2	Level of Analysis	Primary unit of analysis is CEO, TMT, or Board of Directors	Study focuses on middle managers, employees
C3	Construct Focus	Focal construct (narcissism, overconfidence, or hubris) is the primary independent variable	Focal construct appears only as a control, or background reference
C4	Theoretical	Paper engages with UET or	Purely psychological

	Framework	links executive psychological characteristics to organizational outcomes	study with no organizational or strategic outcome
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Table 2 Screening Results

Construct	WoS Initial Retrieval	Post-C1 (AJG ≥ 3)	Excluded (C2–C4)	Final Included
Narcissism	635	290	202	88
Overconfidence	644	286	200	86
Hubris	140	66	35	31
Total	1419	642	437	205

We screened titles and abstracts of all 642 candidate articles after conducting C1. Where the abstract alone was insufficient to determine relevance, we advanced the paper to full-text reading. Applying the inclusion criteria excluded 437 studies, leaving 205 articles for data extraction and synthesis. The final corpus comprised 88 studies on narcissism, 86 on overconfidence, and 31 on hubris. Because some studies ($n = 5$) empirically test two of the three focal constructs as independent variables within the same design, these studies

contribute to both relevant construct counts in Table 2. N therefore reflects construct–study instances rather than strictly unique papers.

To supplement the 205 studies, three additional categories of references were incorporated. First, we conducted forward and backward reference tracking from several foundational studies, including Chatterjee and Hambrick (2007) on CEO narcissism and Malmendier and Tate (2005, 2008) on CEO overconfidence, to capture key contributions that the database searches may have missed. Second, the review draws on foundational theoretical and conceptual works that fall outside the systematic search scope by design, including upper echelons theory (Hambrick & Mason, 1984; Hambrick, 2007), agency theory (Fama & Jensen, 1983; Jensen & Meckling, 1976), stakeholder theory (Freeman, 1984), and foundational construct definitions from psychology (Hare, 1996; Kernberg, 1975; Paulhus & Williams, 2002). These works are necessary to frame the integrative argument but are not themselves empirical studies of the three focal constructs. Third, a small number of empirical studies from adjacent accounting literatures (Ahmed & Duellman, 2013; Hribar & Yang, 2016) are cited where they supply the most direct available evidence on specific outcomes. These studies meet the AJG and construct-focus criteria but were not captured by the search because of disciplinary keyword boundaries, and neither is part of the 205-study evidence base. They serve only to contextualize and interpret the findings.

Data Collection. We created a coding matrix for the 205 included studies. The coding scheme had five dimensions. The first captured research design and sample coverage. This helped us see where the literature had concentrated its empirical efforts and where it had not. For instance, Tang, Li, and Yang (2015) draw on both Eastern and Western contexts, bridging the limitations of prior single-setting studies. The second dimension captured construct operationalization and measurement. This was the most important dimension for our purpose

because the overlaps and divergences in how narcissism, hubris, and overconfidence are measured are a primary source of fragmentation. The third dimension recorded dependent variables and outcome domains. This allowed us to map each construct's empirical footprint: which strategic domains it had been connected to and where evidence remained absent. The fourth dimension recorded the direction of core findings and key moderating conditions. This provided the material for the cross-construct comparisons in Chapter 3. The fifth dimension captured explicitly stated limitations and research gaps, which informed the research agenda in Chapter 5.

Synthesis Approach. The synthesis followed two stages. In the first stage, we analyzed each construct stream separately. We assigned each paper to one or more strategic outcome domains based on the dependent variables coded in the matrix. These domains correspond to the sections in Chapter 3: risk-taking and investment decisions, non-market strategy, financial reporting and ethical behavior, and firm performance. This step revealed each construct's empirical coverage which domains it had addressed and what direction of effect had been most consistently reported (positive, negative, or contingent). In the second stage, we compared the three construct streams side by side within each strategic domain. Where two or more constructs converge on the same domain, we examined whether they produce similar or divergent organizational consequences and traced divergences of any differences back to their definitions, mechanisms, and measurement approaches. This comparison revealed a consistent pattern: the three constructs share a common theoretical root in executive self-enhancement but diverge in the strategic pathways through which self-enhancement shapes organizational outcomes.

3. Results

3.1 Conceptual Foundations and Measurement

Three constructs—narcissism, hubris, and overconfidence—have received substantial attention in upper echelons research, each rooted in a common tendency toward systematic self-inflation: CEOs overestimate their own judgment while discounting external constraints and risk (Chatterjee & Hambrick, 2007; Hayward & Hambrick, 1997; Malmendier & Tate, 2005). Conceptually, they diverge in their defining features: CEO narcissism denotes a personality orientation that craves external admiration and public validation to sustain an inflated self-image (Chatterjee & Hambrick, 2007; Petrenko et al., 2016). CEO hubris refers to a cognitive state characterized by dismissive contempt for evidence that contradicts one's prior decisions (Hayward & Hambrick, 1997; Picone et al., 2014). And CEO overconfidence captures a cognitive bias in which decision-makers systematically overestimate the precision of their own judgments and the probability of successful outcomes (Malmendier & Tate, 2005; Moore & Healy, 2008). This shared tendency has encouraged conceptual conflation across the literature. Sauerwald and Su (2019), for instance, invoke hubris theory to explain the CSR behavior of overconfident CEOs. The practice is far from idiosyncratic and reflects a broader pattern in upper echelons research of substituting one construct for another. Yet a shared inflationary bias does not imply shared strategic consequences. The three constructs differ fundamentally in psychological motivation, information-processing logic, and strategic implications: narcissistic CEOs pursue high-visibility activities to secure external validation and sustain their self-image (Chatterjee & Hambrick, 2007; Petrenko et al., 2016); hubristic CEOs systematically disregard disconfirming information to preserve the validity of their own judgment (Hayward & Hambrick, 1997; Picone et al., 2014); overconfident CEOs drive

high-risk choices by systematically overestimating the probability of success and underestimating downside risk (Malmendier & Tate, 2005; Moore & Healy, 2008). Conflating these constructs obscures the heterogeneous mechanisms underlying CEO strategic behavior and impedes the design of targeted governance interventions (Picone et al., 2014; Tang et al., 2018). This section reviews and compares the conceptual origins and measurement approaches of each construct, establishing the foundation for the cross-construct comparative analyses developed in subsequent sections.

Narcissism

Narcissistic CEOs are fundamentally driven by the pursuit of narcissistic supply (Kernberg, 1975): an ongoing need for external admiration and validation to sustain an inflated self-image. This need shapes how narcissistic CEOs approach strategic decisions: activities that generate an audience are actively pursued; those that do not are neglected. Narcissistic CEOs accordingly show a systematic preference for high-visibility strategic actions while underestimating attendant downside risks and selectively disregarding stakeholder concerns (Chatterjee & Hambrick, 2007; Petrenko et al., 2016).

Measurement of CEO narcissism in upper echelons research has relied predominantly on unobtrusive measures (Chatterjee & Hambrick, 2007). Chatterjee and Hambrick (2007) introduced a composite narcissism index comprising five indicators: the prominence of the CEO's photograph in the annual report, the frequency of CEO mentions in company press releases, the use of first-person singular pronouns in CEO interviews, and the CEO's cash and non-cash compensation relative to the second-highest-paid executive. Peterson et al. (2012) pursued a more direct approach using the NPI-16 self-report scale to reduce the measurement noise inherent in proxy indicators. However, restricted access to CEO samples has limited the broader applicability of this method (Cragun et al., 2020). Notably, Chatterjee and

Hambrick's (2007) stability tests show that narcissism scores for the same CEO are highly correlated across time ($r = .67$ over six years) and across firms ($r = .90$ across job changes). Narcissism operates as a stable personality trait that predates any organizational role rather than being shaped by situational or organizational experience. Complementary linguistic analyses of CEO public communications offer an additional measurement pathway (Craig & Amernic, 2011).

Hubris

The core mechanism of hubris is the active exclusion of disconfirming information from the CEO's decision-making frame. Picone et al. (2014) characterize this as arrogance and overconfidence that prevent an individual from understanding or accepting human limits. This psychological orientation manifests along three dimensions: the systematic discounting of inconsistent evidence, the dismissal and avoidance of others' assessments, and an almost fatalistic sense of personal mission (Picone et al., 2014). Critically, this is not a stable personality trait but a cognitively triggered state that accumulates under specific organizational conditions (Hayward & Hambrick, 1997; Picone et al., 2014). Unlike overconfidence, hubris emerges only in positions of power, reinforced through social interactions that gradually detach the CEO from external constraints (Petit & Bollaert, 2012). This power context is not an abstract feature of the CEO's role. It is built and sustained by a distinct set of external signals. When CEOs experience sustained performance success, receive intensive media praise, and enjoy compensation substantially above that of other executives, the continuous flow of confirmatory feedback progressively erodes the perceived credibility of disconfirming information (Hayward & Hambrick, 1997). In other words, the power context is the very configuration of external signals: organizational success, public

recognition, and social validation. Together, they deepen the CEO's conviction in their own judgment and progressively insulate decision-making from external inputs.

Hubris resists direct measurement because it is a dynamically triggered cognitive state rather than a stable, consistently trackable personality trait. The dominant measurement approach was established by Hayward and Hambrick (1997). They constructed a factor index from three antecedent conditions: recent firm performance, the frequency of favorable media coverage received by the CEO, and the CEO's pay premium relative to other executives. The more fully these triggering conditions are satisfied, the more likely hubris has taken hold. Li and Tang (2010) shifted focus to cognitive outcomes, operationalizing hubris as the industry-adjusted gap between the CEO's subjective performance evaluation and the firm's objective ROA, directly capturing the divergence between self-assessed and externally verifiable judgment. Subsequent research has largely extended or combined these two approaches without departing from their underlying logic (Arena et al., 2018; Tang, Li, & Yang, 2015). Akstinaite et al. (2020) pursued a linguistic strategy, finding that hubristic CEOs' speech is characterized by a marked absence of positive emotion words alongside a co-occurrence of negation and certainty markers.

Overconfidence

Across the literature on CEO overconfidence and strategic decision-making, the construct converges on a single underlying cognitive mechanism: probability miscalibration (Malmendier & Tate, 2005; Moore & Healy, 2008). Moore and Healy (2008) decompose this miscalibration into three mutually reinforcing biases: overestimation, which leads CEOs to inflate assessments of their own capabilities and the likelihood of decision success; overprecision, which produces unwarranted certainty in the accuracy of one's own beliefs; and overplacement, which generates the conviction that one's judgment is superior to that of

peers. Together, these three biases produce a persistent gap between the CEO's subjective probability estimates and external benchmarks, not because information is rejected, but because information, once processed, consistently yields inflated conclusions. This systematic upward bias is not innate. It is acquired through self-attribution: CEOs attribute past decision successes to personal ability rather than situational luck, and each positive outcome further reinforces confidence in the accuracy of their own judgment, deepening probability miscalibration as decision experience accumulates (Billett & Qian, 2008). At the strategic level, this mechanism drives CEOs to systematically accept elevated risk, from premium-priced acquisitions to aggressive R&D commitments. The driver is not a refusal to consider failure, but a persistent belief that their probability of success exceeds what the evidence warrants (Billett & Qian, 2008; Galasso & Simcoe, 2011; Schumacher et al., 2020).

The most widely adopted proxy for CEO overconfidence is whether and when CEOs choose to exercise their stock options (Malmendier & Tate, 2005): A CEO who holds fully vested stock options well above the exercise price rather than exercising them signals a belief that the stock price will keep rising, implying optimism that outstrips rational expectations (Billett & Qian, 2008; Galasso & Simcoe, 2011). This measure embeds an important theoretical assumption: once formed, overconfidence tends to persist, as CEOs do not meaningfully update their subjective probability assessments in response to negative feedback, a condition Galasso and Simcoe (2011, p. 1480) term an "absorbing state". Schumacher et al. (2020) incorporated both the option-based measure (Malmendier & Tate, 2005) and the media-based measure (Malmendier & Tate, 2008) within a single study, using convergence across the two indicators to enhance measurement robustness. The media-based measure proxies overconfidence through the frequency with which financial media attribute firm performance to the CEO personally. The more concentrated this external attribution, the deeper the CEO's self-serving perceptual bias (Malmendier & Tate, 2008).

Hubris and overconfidence are the most commonly conflated pair in upper echelons research, and the confusion is understandable: both manifest as CEO persistence in the face of unfavorable information, and both are positively associated with high-risk strategic behavior. The distinction lies in where the information-processing breakdown occurs. In hubris, the breakdown is prior to processing: CEOs actively mark disconfirming evidence as lacking credibility, such that it never enters the decision-making frame (Hayward & Hambrick, 1997; Picone et al., 2014). In overconfidence, the breakdown occurs during processing: CEOs receive information but systematically weight it toward optimistic conclusions, producing inflated probability estimates (Malmendier & Tate, 2005; Moore & Healy, 2008). This distinction shapes how each construct responds to negative feedback. Hubristic CEOs do not update, because disconfirming information never clears their cognitive filter. Overconfident CEOs are in principle capable of updating, but self-attribution bias continuously reinforces their subjective probability judgments, making meaningful revision equally unlikely in practice (Billett & Qian, 2008; Galasso & Simcoe, 2011). More broadly, all three constructs share a common cognitive root but differ fundamentally in core mechanism and measurement logic: narcissism centers on the relentless pursuit of external validation and is measured through observable self-promotional behavior. Hubris centers on the systematic rejection of disconfirming information and can only be captured through indirect proxies of its antecedent conditions. Overconfidence centers on systematic probability overestimation and is measured through option-exercise behavior that reveals subjective judgment distortions. Treating these constructs as interchangeable means designing governance tools for the wrong mechanism.

3.2 Strategic Risk-Taking and Investment Decisions

All three constructs are positively associated with elevated strategic risk-taking, yet their driving mechanisms are fundamentally distinct. Narcissistic CEOs take risks to create an

audience. Overconfident CEOs take risks because they systematically underestimate the probability of failure, and hubristic CEOs take risks because they disregard the credibility of risk signals.

Narcissism

The strategic choices of narcissistic CEOs follow a stable cognitive rule: the greater an activity's potential visibility and the more readily it can be framed as the conduct of a bold and visionary leader, the more worthy of pursuit it becomes, irrespective of its underlying strategic merit (Chatterjee & Hambrick, 2007; Petrenko et al., 2016). The pursuit of narcissistic supply is the core motivational force animating narcissistic CEOs: they require external recognition to sustain their self-image (Chatterjee & Hambrick, 2007; Kernberg, 1975). Ocasio's (1997) attention-based view provides a further account of how this strategic orientation operates: the CEO's finite attentional resources are directed toward activities that promise the greatest external recognition. Under this logic, high-visibility strategic moves, including innovation, brand acquisition, large-scale mergers, and international expansion, become the natural priority for narcissistic CEOs.

In the innovation decisions domain, narcissistic CEOs systematically display higher innovation activity, but this behavioral pattern is sharply moderated by visibility conditions. When media attention is high, narcissistic CEOs are nearly three times more likely to adopt technology discontinuities than comparable CEOs and when media attention is low, the difference between them becomes negligible (Gerstner et al., 2013). The narcissistic CEO's pursuit of innovation is consistently conditional: the trigger is audience engagement, not the appeal of the technology frontier. This logic is particularly pronounced in the domain of radical innovation: radical digital innovation generates greater social visibility and external

attention, and narcissistic CEOs accordingly invest significantly more heavily in this dimension than their peers (Lee et al., 2025).

In the expansionary strategy domain, both international diversification and brand acquisition extend the firm's operational footprint in ways that continuously amplify the CEO's public exposure and social influence. Along the internationalization dimension, narcissistic CEOs drive the overall growth of the firm's degree of internationalization, but they do not show a systematic tendency to enter countries with high psychic distance, which carry higher risk without necessarily producing stronger audience effects (Oesterle et al., 2016). This strategic behavior is therefore not driven by an active preference for elevated risk. A greater degree of internationalization translates into broader public recognition and a continuous flow of reputational returns associated with being perceived as a globally oriented leader. Along the brand acquisition dimension, narcissistic CEOs are more likely to acquire brands than their non-narcissistic counterparts by 13 percentage points, and among the brands they pursue, they systematically prefer those with high brand awareness (Cao & Xu, 2022). A brand constitutes a reservoir of public attention. Acquiring a brand with high brand awareness is equivalent to acquiring an established audience.

Narcissistic CEOs' risk-taking is not undifferentiated aggression but a highly directional visibility filter: any strategic action capable of attracting audience attention and generating recognition, whether the adoption of technology discontinuities, the advancement of radical digital innovation, geographic expansion through internationalization, or the accumulation of high-awareness brands through acquisition, enters the narcissistic CEO's priority set. Conversely, strategic options that offer more certain economic returns but lack external audience effects are systematically underweighted or neglected.

Hubris and Overconfidence

In the domain of risk-taking, the empirical findings on hubris and overconfidence display a pronounced convergence: both constructs are associated with high-premium acquisitions, aggressive innovation, and over-investment (Hayward & Hambrick, 1997; Galasso & Simcoe, 2011; Tang, Li, et al., 2015). At the level of mechanistic description, the language used in the two research traditions overlaps considerably. Formulations such as overestimating one's capabilities and underestimating the probability of failure appear throughout the theoretical accounts of both constructs, and Picone et al. (2014) note directly that hubris and overconfidence are frequently used interchangeably as synonyms in the finance and management literature. This convergence accounts for why the two research streams cross-cite one another at the empirical level and even share similar theoretical frameworks (Pavićević & Keil, 2021). Yet linguistic overlap is not equivalent to mechanistic convergence, and the differences between the two constructs are most clearly visible at the level of operationalization, which in turn reflects fundamentally different understandings of the nature of each construct.

Hubris measurement is oriented toward its generative sources. Hayward and Hambrick (1997) construct a hubris factor index from three indicators, specifically recent firm performance, favorable media coverage received by the CEO, and the CEO's pay premium relative to other executives. Arena et al. (2018) extend this measurement system by incorporating media coverage, relative compensation, and CEO photograph prominence. The shared characteristic of these indicators is that they measure what feeds hubris: the success signals and recognition signals that the CEO receives from the external environment.

Overconfidence measurement is oriented toward the cognitive bias itself: the Holder67 measure infers the CEO's subjective probability assessment from the decision not to exercise deeply in-the-money stock option grants once fully vested (Malmendier & Tate, 2005; Galasso & Simcoe, 2011). This divergence in measurement logic reveals a difference in

research orientation: hubris researchers ask how external inputs shape CEO confidence, while overconfidence researchers ask how the CEO's internal cognition deviates from objective reality. This distinction does not rest on the language used to describe the mechanism but is grounded in choices made at the level of operationalization.

The differential response of the two constructs to information-based interventions provides the most discriminating empirical evidence. Pavićević & Keil (2021) focus on overconfidence in acquisition decisions and find that when the predeal process is slowed, allowing more thorough due diligence information to enter the decision process, overconfident CEOs' tendency to overpay is significantly attenuated. This finding is highly consistent with the mechanistic prediction for overconfidence: the bias operates at the stage of information processing, where information is received but optimistically revised at the point of probability assessment. When more thorough information enters the decision frame, the calibration of the processing improves and the bias is accordingly reduced. The same logic simultaneously generates a directionally opposite testable prediction for the hubris construct: hubris operates before information enters the decision frame: hubristic CEOs actively deny the credibility of disconfirming evidence. The same slow-decision intervention that corrects overconfidence would therefore fail with hubristic CEOs. Such interventions introduce precisely the kind of information that hubristic CEOs have already excluded. The difference between the two mechanisms in the location of the information-processing breakdown thereby generates a prediction of differential responses to the same intervention. That prediction is directly testable, and the framework developed in this review provides a clear path for future research to verify it.

Hubristic CEOs' innovation behavior displays a different pattern of moderation. Tang, Li, and Yang (2015) find that both environmental munificence and environmental complexity

attenuate hubristic CEOs' innovation investment: in munificent environments, the abundance of strategic opportunities, and in complex environments, the complexity of the competitive landscape, both reduce the CEO's capacity to maintain focus on innovation. This difference carries implications: the moderating factor for overconfidence is the intensity of external incentive signals, where stronger competitive pressure makes the consequences of probability miscalibration more pronounced (Galasso & Simcoe, 2011). The moderating factor for hubris, by contrast, is the density of opportunities and distractions in the environment. Greater munificence or complexity in the environment makes it harder for the CEO to sustain attentional focus on innovation.

The convergence and divergence of hubris and overconfidence in the risk-taking domain provide a key insight into the same-root-different-paths logic: a shared self-enhancement root generates similar surface behaviors across constructs, but the distinction between constructs lies not in what they do but in why they do it and under what conditions the bias admits of correction.

3.3 Non-Market Strategy: CSR and ESG Performance

Research on CEO psycho-social characteristics has increasingly extended beyond financial decisions to examine how executive traits shape non-market strategy. Stakeholder theory establishes that firms depend on the resources, legitimacy, and cooperation of diverse external constituencies (Freeman, 1984), and how CEOs process stakeholder demands will determine whether and how their firms invest in CSR. However, whether narcissistic, hubristic, or overconfident CEOs engage with CSR depends critically on the nature of the CSR activity itself. Porter and Kramer (2006) distinguish between responsive CSR, through which firms address social harms in response to external stakeholder pressure, and proactive CSR, through which firms pursue social initiatives driven by internal conviction. Different

psychological mechanisms generate fundamentally different sensitivities to these two categories of CSR activity. This section traces the divergent effects of narcissism, hubris, and overconfidence on CSR engagement and ESG performance.

Narcissism

Narcissistic CEOs are more likely to engage in CSR than their less narcissistic counterparts, a positive relationship that holds across multiple samples and institutional contexts (Al-Shammari et al., 2019; Petrenko et al., 2016; Tang et al., 2018). The underlying mechanism is not genuine stakeholder orientation but the use of CSR as a vehicle for narcissistic supply (Kernberg, 1975): the external recognition and adulation that narcissistic individuals require to sustain a positive self-view (Chatterjee & Hambrick, 2007, 2011; Petrenko et al., 2016). CSR positions the CEO on visible moral high ground, generates media attention and stakeholder praise that audiences attribute to the CEO personally rather than to the firm, and provides ongoing occasions for public recognition (Petrenko et al., 2016). Because the demand for this supply is continuous, narcissistic CEOs have a persistent incentive to maintain and expand CSR investment.

This mechanism also predicts that narcissistic CEOs will allocate CSR resources selectively on the basis of the attention-base perspective, favoring activities that command large external audiences over those with limited public visibility. CEO narcissism is positively related to externally oriented CSR activities such as community engagement, environmental programs, and charitable giving, but is not significantly related to internally oriented activities such as promotions, hiring equity, equal pay, and employee training (Al-Shammari et al., 2019). The same pattern holds in Eastern context: narcissistic CEOs invest more in peripheral CSR activities that are highly visible and personally attributable to the CEO, and less in embedded CSR that is integrated into core operations and less transparent to outside observers (Chen et al., 2021).

The image-driven logic that underlies narcissistic CSR participation reveals its substantive limits when the analysis shifts to ESG performance. Petrenko et al. (2016) argue that narcissistic CEOs initiate more CSR activities, but this participation does not translate into stable stakeholder value. And CEO narcissism weakens the link between ESG performance and firm value. Even when ESG performance is strong, narcissistic CEOs capture less of the value that performance should generate (Effah et al., 2026). Gloor et al. (2026) supply the mechanism: narcissistic CEOs' environmental performance is contingent on anticipated media rewards rather than stakeholder commitment, and declines when positive media coverage is withdrawn. Narcissistic supply-seeking produces a systematic gap between CSR quantity and ESG quality.

Hubris

Hubris generates a CSR profile that is opposite to narcissism in the responsive domain. The evidence contempt mechanism leads hubristic CEOs to discount stakeholder accounts of what the firm owes them, fostering an unwarranted sense of self-sufficiency (Arena et al., 2018; Tang, Qian, et al., 2015): they overestimate the firm's internal resource endowment (Li & Tang, 2010) and underestimate the resources that stakeholder cooperation provides, thereby underestimating how much the firm depends on its external constituencies (Tang, Qian, et al., 2015). Stakeholder demands for product safety, workforce equity, community participation, and employee welfare are therefore treated not as legitimate constraints on firm behavior but as the claims of actors whose evaluations carry less weight than executive judgment. The result is a pattern in which hubris is negatively associated with CSR strengths and positively associated with CSR concerns (Tang, Qian, et al., 2015; Tang et al., 2018).

The same evidence contempt mechanism generates the opposite prediction for proactive CSR. CEO hubris is positively associated with environmental innovation (Arena et al., 2018) and with investment in R&D more broadly (Tang, Li, et al., 2015), even when such

commitments involve high outcome uncertainty and long payback horizons. A CEO who discounts external risk assessments will not be deterred by the unfavorable probability estimates that inhibit most executives from ambitious environmental commitments, and an inflated belief in personal judgment sustains commitment through periods of uncertain return. Responsive CSR requires accepting stakeholders' definitions of what responsible conduct entails, while proactive CSR allows the CEO to set the initiative's scope and standards according to their own conviction. Hubristic CEOs reject externally defined obligations in the former domain while remaining unconstrained in the latter, where evidence contempt does not suppress but instead facilitates internally motivated social innovation.

Overconfidence

CEO overconfidence has attracted comparatively less attention in the non-market strategy literature than the other two constructs, but the existing evidence converges on a characteristic pattern: a gap between CSR reporting and actual performance. Sauerwald and Su (2019) find that overconfident CEOs overestimate their capacity to execute CSR initiatives; when actual performance falls below expectations, they respond by amplifying symbolic CSR reporting to protect an inflated self-image from disconfirming evidence, widening rather than closing the reporting-performance gap. The same miscalibration simultaneously suppresses awareness of the legitimacy costs that stakeholders impose when the gap becomes visible (Sauerwald & Su, 2019). From a symbolic management perspective, this pattern in which overconfident CEOs promote expressive CSR declarations that outpace substantive delivery is negatively associated with CSR quality, with board oversight providing only a partial check.

Overconfidence also shapes environmental performance, the environmental pillar of ESG. Overconfident executives are more likely to violate environmental regulations:

Andrikopoulos et al. (2025) find that CFO overconfidence increases environmental violations, which in turn hurt long-term firm performance. This effect is weaker when firms face stronger stakeholder protection laws or heavier penalties. The reason is straightforward: overconfident executives underestimate the likelihood of detection and the severity of punishment. They treat compliance as an unnecessary expense, because they do not believe negative consequences will happen to them.

Across the three constructs, a pattern of convergent outcomes produced by divergent mechanisms emerges as a structural regularity in the non-market domain. Narcissism, hubris, and overconfidence each generate negative ESG performance, yet through mechanisms that differ in character and the specific CSR domains they affect: narcissistic supply-seeking produces a gap between CSR quantity and ESG quality; evidence contempt lead different CSR engagement by initiative type, amplifying proactive investment while suppressing responsive obligations; probability miscalibration generates CSR decoupling(Sauerwald & Su, 2019).

3.4 Financial Reporting and Ethical Behavior

In the domain of financial reporting, all three constructs can produce information that deviates from an accurate representation of firm performance, yet through mechanisms that differ in both psychology and ethics. One is self-serving misrepresentation, another is an inability to recognize the ethical dimension at all, and the third is good-faith miscalibration.

Narcissism

Narcissistic CEOs are more inclined to treat financial reporting as an instrument for seeking admiration and self-affirmation rather than as a medium for objectively communicating firm performance to stakeholders. This inclination originates in the psychological projection

through which narcissistic CEOs equate firm performance with personal capability (Amernic & Craig, 2010), and the cyclical structure of financial reporting aligns naturally with their persistent demand for external validation (Rijssenbilt & Commandeur, 2013). Accounting is widely perceived in social cognition as a domain of fairness and objectivity, a quality that Amernic and Craig (2010, p. 86) term the "aura of accounting." This cultural authority enables narcissistic CEOs to borrow the legitimacy of accounting to ratify their own conduct: once manipulated figures pass audit and receive the imprimatur of fair presentation, the narcissistic CEO can present fabricated performance as externally verified fact, constructing within public discourse a self-image that is decoupled from actual results.

These theoretical propositions receive direct empirical support from Rijssenbilt and Commandeur (2013), who analyzed a sample of 953 S&P 500 CEOs over the period 1992 to 2008 and found that narcissistic CEOs are significantly more likely to be implicated in financial fraud, which is an ethical violation. Among the 54 CEOs charged with or confirmed as having engaged in fraudulent conduct by the Securities and Exchange Commission, narcissism scores were significantly elevated. The underlying mechanism is that narcissistic CEOs set unrealistically high performance targets, and when those targets cannot be met, they revise financial reporting data rather than acknowledge failure. For narcissistic CEOs, acknowledging failure implies acknowledging inadequacy, an outcome their cognitive structure renders inadmissible, and revising reported figures is both faster and more direct than improving actual performance.

When the degree of manipulation remains below the threshold of fraud, the same narcissistic supply logic operates through day-to-day earnings management. Buchholz et al. (2020) find that CEO narcissism is significantly and positively associated with accrual-based earnings management (ABEM), realized through two opposing mechanisms. The first is subconscious

bias: cognitive distortion leads narcissistic CEOs to overestimate their own capabilities and firm performance, inflating current-period earnings without deliberate intent. The second is deliberate bias: narcissistic CEOs consciously suppress current-period earnings, particularly around CEO succession, as incoming narcissistic CEOs deflect performance pressure onto their predecessors through income-decreasing earnings management while simultaneously reserving headroom for future earnings growth. Across both mechanisms, the common objective is identical regardless of direction: earnings management serves to preserve the integrity of the self-image. When earnings management suffices to sustain that image, narcissistic CEOs remain within the domain of legal but low-quality accounting; when it can no longer conceal genuine performance deterioration, crossing the fraud threshold becomes the extreme extension of the same underlying logic.

Hubris

The pathway through which hubristic CEOs produce financial reporting misconduct differs fundamentally from that of narcissism. Where narcissistic CEOs instrumentalize accounting as a medium of self-verification through deliberate manipulation, the fundamental mechanism underlying financial distortion in hubristic CEOs is the impairment of moral awareness (McManus, 2018). Moral awareness, defined as the capacity to recognize that a decision contains an ethical dimension, is a prerequisite for ethical decision-making. When this prerequisite is compromised, CEOs frame decisions as business problems rather than ethical ones, and attention shifts from whether an action conforms to rules to how the objective can be achieved. The practical consequence of this reframing is that accounting rules and disclosure requirements, as a category of external information, are excluded from the CEO's decision calculus. They are not evaluated and rejected; they are never admitted to consideration. McManus (2018) provides direct empirical support for this mechanism: hubris

proxies constructed from media praise, CEO relative compensation, and biographical length all significantly predict earnings restatement.

Moral awareness impairment operates through two pathways. The first is an internal orientation bias: hubris focuses the CEO's attention on internal factors, including personal capability, judgment, and prior success, while suppressing sensitivity to external situational cues such as the severity of decision consequences, the breadth of affected parties, and prevailing social norms (McManus, 2018). The second is a shift in reasoning style: hubris disposes CEOs toward utilitarian reasoning, in which the anticipated success of an action is taken as evidence of its rightness, rather than formalist reasoning, in which conformity to rules defines what is right. Reynolds (2006) demonstrates that formalist reasoning is more likely than utilitarian reasoning to activate moral awareness. Both pathways converge on a common outcome: external information, whether risk signals, social norms, or formal rules, is excluded before it can enter the decision.

The origin of this cognitive pattern lies in the relationship that hubristic CEOs form with the world beyond the firm. Petit and Bollaert (2012) identify three cognitive dimensions of hubris: a grandiose self-conception, an instrumental orientation toward others, and a contempt for the established order of the world. It is the third dimension, a systematic disregard for rules and laws, that leads hubristic CEOs under performance pressure to treat accounting requirements as external constraints unworthy of serious consideration.

Overconfidence

Unlike the strategic manipulation of accounting figures that characterizes narcissism and the selective filtering of discrepant evidence that characterizes hubris, the distortion of financial reporting produced by CEO overconfidence is rooted in probability miscalibration.

Overconfident CEOs do not intend to inflate figures; they genuinely believe that their

optimistic assessments of the future are accurate and record those assessments faithfully in their financial statements. This cognitive bias operates across the full range of financial decisions requiring subjective judgment, from the routine valuation of assets and recognition of losses (Ahmed & Duellman, 2013) to voluntary management earnings forecasts (Hribar & Yang, 2016), and ultimately to a gradual slide toward financial misreporting under performance pressure (Schrand & Zechman, 2012).

Ahmed and Duellman (2013) find that the accounting policies of overconfident CEOs are systematically biased toward optimism across two dimensions. First, when projects show signs of loss, overconfident CEOs tend to defer recognition, proceeding on the assumption that the loss is temporary and that future performance will recover. Second, in the routine valuation of assets, overconfident CEOs consistently overestimate: they overstate the recoverability of receivables and the remaining useful lives of fixed assets, producing systematically inflated book values. The cumulative effect of this sustained micro-level optimism is that the financial reports of overconfident CEOs contain a systematic initial overstatement at their source (Schrand & Zechman, 2012).

The same cognitive disposition shapes how overconfident CEOs communicate with the market. Hribar and Yang (2016) find that overconfident CEOs are more likely to issue voluntary earnings forecasts, and that those forecasts are systematically optimistic, setting targets that prove difficult to achieve. When accounting-level initial overstatements accumulate alongside market-level forecasts that cannot be met, the conditions that Schrand and Zechman (2012) characterize as a slippery slope emerge. Overconfident CEOs, continuing to believe that next period's performance will recover, do not acknowledge original overstatements but instead make progressively harder-to-reverse adjustments under mounting pressure, eventually sliding into active misreporting. In this trajectory, the starting

point is not deception but an excess of confidence in one's own judgment. This is ethically distinct from the other two constructs: it is closer to negligence than to fraud.

3.5 Firm Performance Consequences

Unlike the direct effects that the three constructs exert in the domains examined in previous sections, firm performance does not arise as an immediate consequence of CEO psycho-social characteristics. Rather, each construct shapes performance by altering the strategic behaviors through which managerial agency is exercised, and the mediating strategic behaviors generated by distinct psychological mechanisms produce correspondingly distinct performance profiles. This section traces how narcissism, hubris, and overconfidence each mediate firm performance through characteristic strategic pathways, and identifies how different psychological mechanisms translate a shared self-enhancement root into fundamentally differentiated performance outcomes. Studies in this domain measure performance in different ways. Some use accounting figures like ROA and ROE. Others use market-based indicators such as TSR or acquisition premiums. Still others rely on meta-analytic summaries. Rather than comparing the specific numbers these measures produce, this review identifies patterns that appear consistently within each construct. If a pattern holds across different types of measures, it is treated as reflecting the construct itself, not the way it was measured.

Narcissism

CEO narcissism produces high performance variance rather than consistent outperformance or underperformance (Chatterjee & Hambrick, 2007; Wales et al., 2013). This high variance appears consistently across different measures. It holds whether performance is tracked through return on assets (Chatterjee & Hambrick, 2007) or through archival sales data (Wales et al., 2013). The mediating mechanism lies in the distinctive selection logic that governs the

strategic portfolio of narcissistic CEOs: in seeking external attention and self-validation, narcissistic CEOs disproportionately favor bold, high-visibility strategies, including aggressive innovation (Gerstner et al., 2013; Lee et al., 2025), branded acquisitions (Cao & Xu, 2022), international expansion (Oesterle et al., 2016), and externally oriented CSR (Al-Shammari et al., 2019; Petrenko et al., 2016). The selection criterion governing this portfolio is not expected economic return but the capacity of each initiative to generate the attention and admiration that constitute narcissistic supply. This logic produces extreme performance variance: high-visibility strategies may coincide with genuinely high-return projects and drive superior performance, or they may constitute costly spectacles that consume resources without generating substantive economic value and produce underperformance. Firms led by narcissistic CEOs do not, therefore, systematically outperform or underperform their peers in aggregate (Chatterjee & Hambrick, 2007). Kraft's (2022) meta-analysis of 68 studies supplies the mechanism underlying this pattern of nonsignificant mean effects alongside substantial variance: narcissistic CEOs generate a positive indirect performance effect through their promotion of innovation, while simultaneously producing a significant negative direct effect through other strategic behaviors; these opposing pathways coexist and offset each other, driving the net effect toward zero.

Hubris

In contrast to the extreme performance variance associated with CEO narcissism, hubristic CEOs generate performance outcomes that are consistently negative and resistant to self-correction. Evidence contempt leads hubristic CEOs to actively reject information that is inconsistent with their existing beliefs (Picone et al., 2014), preventing risk signals, failure indicators, and negative feedback from entering the decision-making framework.

Consequently, even when initial strategic behaviors produce negative performance outcomes, the absence of a correction mechanism allows decision errors to accumulate and risk exposures to compound. Picone et al. (2014) conceptualize this dynamic as the hubris trap: success reinforces hubris, hubris generates strategic miscalculation, the CEO refuses to acknowledge failure and instead launches a more ambitious initiative to vindicate the original judgment, producing even greater failure. The cumulative effect is a downward spiral from which hubristic CEOs cannot recover, precisely because recovery would require acknowledging the errors that evidence contempt compels them to deny. Hayward and Hambrick (1997) provide empirical confirmation of this mechanism in the acquisition domain: hubristic CEOs systematically disregard target valuation signals and integration risks, paying acquisition premiums that increase with the depth of their hubris, ultimately imposing large wealth losses on acquiring firm shareholders. Park et al. (2018) extend the evidence beyond the acquisition context, using return on assets relative to the industry median as the performance measure: controlling for firm size, industry, and governance structure, CEO hubris retains a significant negative main effect on firm performance, demonstrating that evidence contempt is not confined to high-stakes acquisition decisions but permeates strategic decision-making broadly. The firm performance trajectory of hubristic CEOs is therefore one of sustained negative deviation rather than the high-variance pattern characteristic of narcissistic leadership.

Overconfidence

The influence of CEO overconfidence on firm performance originates in a cognitive mechanism that is fundamentally distinct from hubris. Where evidence contempt operates by filtering information before it enters the decision framework, overconfidence operates through probability miscalibration (Moore & Healy, 2008): overconfident CEOs do not reject

incoming information but systematically overestimate the probability of success and underestimate the probability of failure when evaluating it. This miscalibration alters the risk-return calculations that govern strategic choice, disposing overconfident CEOs toward higher-risk options including expanded R&D investment, elevated capital expenditure, and entry into markets characterized by greater uncertainty. In favorable economic conditions, this strategic aggressiveness can generate genuine performance premiums. Burkhard et al. (2023) synthesized 199 studies and found a small but consistent positive effect. This pattern holds across both accounting-based returns, such as ROA and ROE, and market-based performance indicators. The positive effect is, however, context-dependent. Reyes et al. (2022) find that overconfident CEOs generate return on equity 1.86 percentage points above industry peers during economic expansions, but this advantage disappears entirely during downturns: the aggressive investment and accelerated decision-making that aligns with rising market conditions produces organizational rigidity through prior asset accumulation, leaving these firms unable to adapt when conditions deteriorate. This pattern of swinging sharply with economic context constitutes the substantive content of high performance volatility under overconfident leadership. Taken together with the positive mean effect documented by Burkhard et al. (2023), the performance profile of overconfident CEOs is one of above-average performance on average accompanied by substantially higher outcome dispersion relative to peers. Unlike hubris, probability miscalibration is correctable when decision-makers are exposed to sufficient information: Pavićević & Keil (2021) find that when acquisition decisions are deliberately slowed and subjected to more thorough due diligence, the tendency of overconfident CEOs to pay excessive premiums is significantly attenuated. Relative to narcissism and hubris, CEO overconfidence is the only one of the three constructs to generate a positive performance mean, and the only psychological bias among the three that is amenable to informational correction.

4. Discussion

4.1 Toward a Cross-Construct Integration

Narcissism, hubris, and overconfidence all originate in a common cognitive root manifesting as a tendency to overestimate personal capability and underestimate external risk (Chatterjee & Hambrick, 2007; Hayward & Hambrick, 1997; Malmendier & Tate, 2005). A shared root does not, however, produce shared strategic consequences. The core distinction among the three constructs lies in how each sustains self-enhancement bias: narcissistic CEOs seek external praise and attention; hubristic CEOs exclude information inconsistent with their existing beliefs; overconfident CEOs overestimate the accuracy of their own judgment. Each construct converts the common self-enhancement drive into differentiated strategic behavior through a distinct pathway. These distinct pathways reflect differences in how the three constructs originate. Narcissism is a stable personality trait that predates the CEO role; the self-enhancement need is persistent and is repeatedly confirmed through attention-seeking. Hubris is a state-dependent bias cultivated by power and success; accumulated recognition gradually erodes the CEO's confidence in external information, leading them to reject disconfirming signals in order to preserve existing judgments. Overconfidence originates in self-serving attribution: the persistent tendency to attribute successes to personal ability and failures to external circumstances inflates subjective confidence in the accuracy of one's own assessments, producing systematic errors in probability calibration.

These distinct pathways correspond to different stages at which bias enters the decision-making chain. Attention allocation, information intake and information evaluation together form the complete sequence through which any strategic decision must pass. The bias of narcissistic CEOs operates at the attention allocation stage: attention and resources are

directed toward activities that generate high audience visibility, while information that cannot produce such attention is deprioritized. Options entering the narcissistic CEO's consideration set are ranked by visibility rather than expected return, with high-visibility alternatives systematically preferred. Gerstner et al. (2013) found that narcissistic CEOs are nearly three times more likely to adopt technological discontinuities when media attention is high, but this difference disappears entirely when attention is low. The vanishing of the narcissism effect under low-visibility conditions is precisely what an attention allocation stage filter predicts: a strategy that offers no visibility does not register as worth pursuing and never enters consideration. The bias of hubristic CEOs operates at the information intake stage: risk signals and failure indicators are filtered before they can enter the decision frame, leaving strategic deliberation populated exclusively by information consistent with prior convictions. Kowalick and Appels (2023) provide direct empirical support for this. More hubristic CEOs undertake fewer strategic changes and less TMT membership change, even when external signals suggest a need for adjustment. The signals that would prompt strategic reconsideration are the kind of disconfirming information that evidence contempt filters out. Because these signals challenge the CEO's existing beliefs about the adequacy of current strategy, they are rejected before they can influence decision-making. The bias of overconfident CEOs operates at the information evaluation stage: information is permitted to enter the decision frame, but is assessed in a manner that systematically overestimates success probabilities and discounts downside risk. Pavićević & Keil (2021) found that when acquisition decisions are deliberately slowed and more thorough due diligence information is introduced, the tendency of overconfident CEOs to overpay is significantly weakened. If the bias operated at the information intake stage, introducing more information would have no effect. This is precisely why overconfidence belongs at the evaluation stage: the information gets in, but it is processed incorrectly.

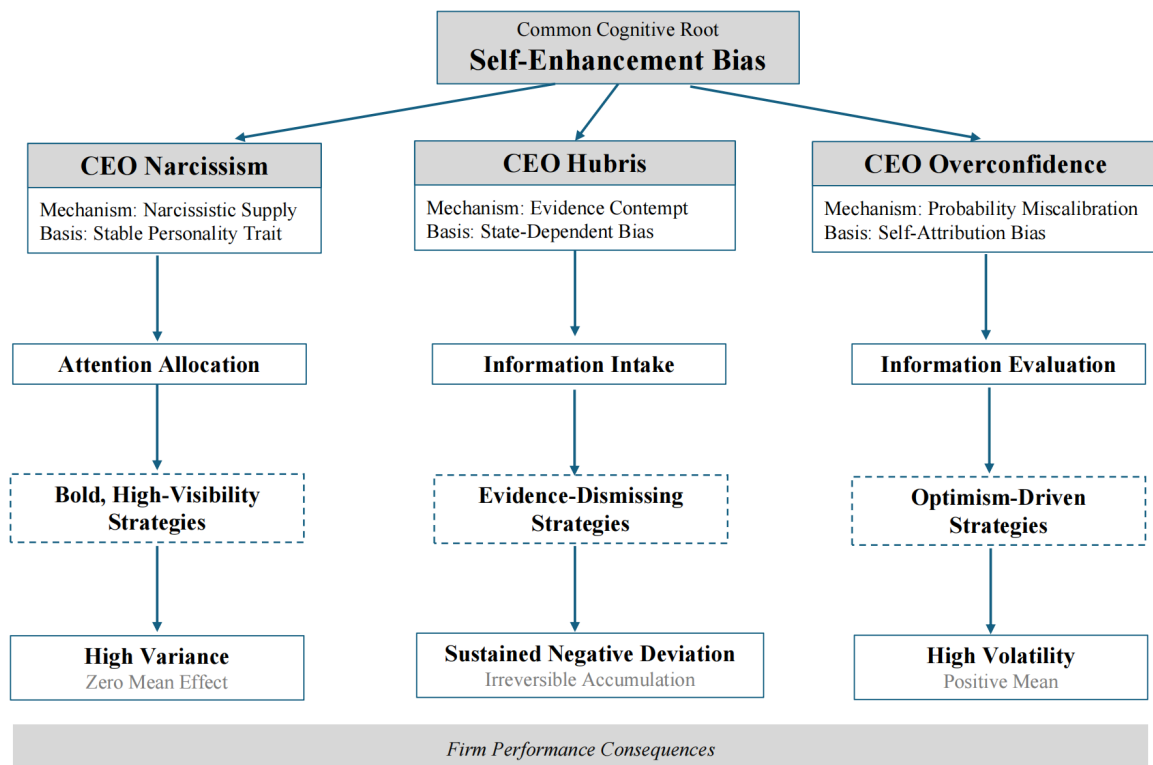


Figure 1. *An Integrative Framework of CEO Narcissism, Hubris, and Overconfidence*

This differentiation in mechanisms means that identical governance structures and market conditions produce different moderating effects across the three constructs, through fundamentally different pathways. Effective boundary conditions must engage the specific stage at which a given construct generates bias rather than applying generalized external pressure. Within internal governance, independent director mechanisms constrain all three constructs to some degree, but through entirely different pathways. For narcissistic CEOs, who favor high-visibility actions, independent directors with industry-specific expertise can identify and challenge strategically inappropriate initiatives, producing some attenuation of risk-seeking behavior (Buyl et al., 2019). This finding also illustrates that directorial oversight cannot eliminate the influence of narcissism but can only redirect it toward less harmful expressions: narcissism, as a stable personality trait, presents a constraint problem at the governance level that behavioral monitoring cannot resolve. For hubristic CEOs, the critical governance variable is whether the board contains voices capable of challenging the

CEO's judgment and compelling engagement with adverse information. CEO duality systematically amplifies the effects of hubris precisely because it severs the channel through which contrary information might otherwise be introduced (Hayward & Hambrick, 1997; Li & Tang, 2010). For overconfident CEOs, the requirement is not more oversight but more specialized oversight: only independent directors with domain-relevant expertise can identify the systematic biases in a CEO's probability assessments, while generalist monitoring experience produces substantially weaker effects (Sauerwald & Su, 2019).

External market conditions have been extensively studied as boundary conditions in this literature. The integrative analysis identifies a structural regularity: market uncertainty and competitive intensity operate in opposite directions across constructs. Market uncertainty attenuates the negative effects of CEO hubris (Tang, Qian, et al., 2015): under high uncertainty and competitive pressure, firms become more dependent on external stakeholders, compelling CEOs to engage with a broader range of external information and thereby partially counteracting the information filtering characteristic of hubristic decision-making. Competitive pressure amplifies the negative effects of CEO overconfidence, however (Galasso & Simcoe, 2011): intense competition reduces strategic information disclosure across firms, leaving overconfident CEOs without the external reference points needed to calibrate probability estimates, and subjective overestimation intensifies accordingly.

These differential boundary conditions carry direct implications for the design of governance interventions. Because narcissistic CEOs' behavior follows the direction of attention rather than any particular strategic category, acquisitions, externally oriented CSR, and aggressive innovation are interchangeable attention sources; restricting any one category will produce substitution toward another high-visibility alternative, leaving the underlying drive unaffected. Effective intervention must therefore operate at the level of the attention

allocation mechanism itself rather than at the level of specific behavioral expressions. For hubristic CEOs, whose bias operates at the information intake stage, effective intervention requires the external introduction of information that the CEO's filtering mechanism would otherwise exclude. For overconfident CEOs, whose bias operates at the information evaluation stage, effective intervention requires providing richer informational inputs against which subjective probability estimates can be calibrated.

4.2 Theoretical Contributions

This integrative review advances existing theory along three dimensions. The first contribution is the construction of a unified Same Root, Divergent Forms framework that simultaneously incorporates CEO narcissism, hubris, and overconfidence. Earlier reviews have examined subsets or broader sets of executive psycho-social constructs (Borgholthaus et al., 2023; Brunzel, 2021). And Heavey et al. (2022) position executive confidence as a central construct with hubris as one of its manifestations. But none has systematically compared these three self-enhancement constructs side by side to identify what they share, how they diverge, and what those differences imply for strategic behavior and governance. The framework fills this gap by establishing two points. First, all three constructs share the same origin. Second, they differ in where bias occurs. The framework thus provides, within a single analytical structure, an account of both why the three constructs resemble one another and why they produce divergent strategic effects. By integrating empirical findings distributed across separate research streams, the framework advances the broader movement of the field from construct-isolated inquiry toward cross-construct comparative research.

The second contribution is a more fine-grained microfoundational account of upper echelons theory (Hambrick & Mason, 1984). Upper echelons theory established the foundational proposition that CEO psycho-social characteristics shape strategic decision-making. The

theory offers a research perspective rather than a unified theoretical account, however, and provides a relatively limited treatment of the intermediate cognitive pathways through which psycho-social characteristics translate into strategic outcomes (Bromiley & Rau, 2016). By tracing the divergent effects of the three constructs to specific stages of the information-processing chain, this review advances the upper echelons analytic from a trait-to-outcome model to a model in which the pathway runs from trait to cognitive fault point to strategic decision to organizational outcome.

The third contribution is the identification of construct specificity in effective governance. The corporate governance literature has tended to treat mechanisms such as independent directors and the separation of CEO and board chair roles as general-purpose instruments for constraining CEO behavioral bias (Fama & Jensen, 1983), without systematically distinguishing how different CEO characteristics respond to identical governance structures. The systematic cross-construct comparison conducted in this review demonstrates that identical governance mechanisms produce fundamentally different moderating effects across constructs. Whether a given mechanism is effective depends on whether it engages the cognitive stage at which the target construct generates bias.

4.3 Governance Implications

Agency theory frames corporate governance around two core mechanisms: board monitoring to constrain managerial opportunism, and equity-based compensation to align managerial interests with those of shareholders (Jensen & Meckling, 1976). The majority of research on CEO governance is built on this framework. The integrative framework developed in this review surfaces a question that agency theory has not yet adequately addressed: when CEO cognitive bias originates at different stages of the information-processing chain, the effectiveness of monitoring and incentive mechanisms varies systematically across

constructs. This section traces the governance logic implied by each construct along both pathways.

Narcissistic CEOs direct resources toward strategies capable of generating attention and recognition, and restricting any particular high-visibility behavior produces substitution toward an alternative. Post-hoc monitoring of specific strategic behaviors is therefore of limited effectiveness, and intervention must be directed at the attention allocation mechanism itself. On the incentive side, stock option holdings amplify the risk-seeking behavior of narcissistic CEOs because options reinforce the connection between high-visibility strategic action and personal financial reward; narcissistic CEOs already exhibit a preference for such actions, and two motivational forces thus converge on the same behavioral direction (Buyl et al., 2019). The logic is not primarily financial. Compensation metrics signal what the board treats as important; shifting those signals shifts the CEO's attention accordingly.

Compensation committees can address this by restructuring the metric composition of executive incentive plans, reducing the incentive weight assigned to high-visibility, short-term performance indicators and increasing the proportion of long-term value metrics in the overall compensation structure.

The bias of hubristic CEOs operates at the information intake stage. Conventional board monitoring reviews what CEOs present for deliberation, but the content of such presentations has already been filtered; adverse signals are excluded before they reach the board. The board thus observes a curated informational set and is unable to engage the bias at its source.

Effective monitoring must therefore be reconceived: the objective shifts from the passive review of presented decisions to the active introduction of information that the CEO's filtering mechanism would otherwise prevent from reaching deliberation. Maintaining the separation of CEO and board chair roles is a foundational governance requirement in this

context, as CEO duality institutionally severs the channel through which contrary information might otherwise be introduced and has been found to systematically amplify the effects of hubris (Hayward & Hambrick, 1997; Li & Tang, 2010). On the incentive side, the problem of hubristic CEOs does not lie in motivational misalignment, and compensation adjustments are unlikely to alter the underlying behavior. Requiring major strategic decisions to undergo external expert evaluation or multi-level deliberation is a more directly targeted intervention, as it institutionalizes the introduction of perspectives that the CEO's filtering mechanism would otherwise prevent.

The bias of overconfident CEOs operates at the information evaluation stage: information enters the decision frame, but is assessed in a manner that systematically overestimates the probability of success. Board monitoring carries greater traction here than in the case of hubris, because the CEO's probability assessments are observable and can be compared against external benchmarks. The effectiveness of monitoring is highly contingent on the domain expertise of those conducting it, however; only independent directors with field-relevant knowledge can identify and challenge the systematic errors in a CEO's probability judgments, while generalist oversight experience produces substantially weaker effects (Sauerwald & Su, 2019). Board capability development for firms led by overconfident CEOs must therefore prioritize depth of domain expertise over increases in independence ratios or board size. Slowing decision tempo and requiring thorough due diligence cycles additionally provide the external informational inputs against which subjective probability estimates can be calibrated (Pavićević & Keil, 2021).

5. Limitations and Future Research Directions

While our study adds to existing knowledge, it has certain limitations that present opportunities for further investigation. First, any integrative review carries inferential

constraints that single-study empirical work does not. The framework developed in this review rests on theoretical integration across independent empirical studies rather than direct empirical testing within a single research design. The conclusions drawn regarding the differential characteristics of CEO narcissism, hubris, and overconfidence are inferred from independent studies that differ in sample composition, measurement instruments, and temporal coverage. Cross-study inference carries a fundamental limitation: observed differences may reflect sample heterogeneity or methodological heterogeneity rather than genuine construct-level distinctions. Future research should simultaneously incorporate all three constructs within a single empirical study, enabling direct comparisons of their differential effects on strategic decisions and firm performance within a common setting, and thereby eliminating the confounding alternatives that cross-study inference cannot rule out.

Second, the dominant operationalizations of each construct were developed independently and are not designed to enable cross-construct comparison. Narcissism proxies capture stable personality expression through unobtrusive archival indicators (Chatterjee & Hambrick, 2007); hubris proxies capture state-dependent bias arising from past success through archival antecedent indicators (Hayward & Hambrick, 1997); and overconfidence proxies capture the behavioral manifestation of probability miscalibration through option exercise patterns (Malmendier & Tate, 2005). The fundamental differences in measurement logic make it difficult to cleanly attribute observed effect differences in cross-construct comparisons to genuine construct distinctions rather than to differences in measurement approach. Future research should develop measurement instruments or proxy indicators oriented toward diagnosing the specific stage at which executive bias originates, with the objective not of determining which dark personality construct a CEO possesses, but of locating whether the bias operates at the attention allocation, information intake, or probability calibration stage.

A third limitation is that existing research treats the three constructs as mutually exclusive types, implicitly assuming that any given CEO exhibits only one form of bias. Although a small number of studies have examined interactions between conceptually opposed constructs, such as humility and narcissism (Nie et al., 2022; Vianello et al., 2026), the co-occurrence of narcissism, hubris, and overconfidence within the same executive has received virtually no empirical attention. The integrative framework developed in this review additionally predicts that the simultaneous activation of multiple fault points may produce nonlinear effects whose consequences exceed the sum of the individual construct contributions; the presence of one bias may amplify or suppress the behavioral expression of another. Future research should examine whether CEOs who simultaneously exhibit multiple dark personality characteristics produce more extreme strategic and financial outcomes, and whether the pattern of interaction, whether additive, amplifying, or heterogeneous in form, varies systematically with contextual conditions.

6. Conclusion

This review developed the Same Root, Divergent Forms integrative framework, documenting a systematic pattern in which CEO narcissism, hubris, and overconfidence share a common self-enhancement root yet differentiate into three psychologically distinct mechanisms and divergent strategic consequences. The framework advances upper echelons theory by providing a more fine-grained account of the cognitive pathways through which executive characteristics shape organizational outcomes, and offers differentiated theoretical grounding for the design of board-level governance interventions.

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Appendix A

List of Articles Included in the Review

This appendix lists the 205 construct–study instances (200 unique articles; AJG 2024 rating $\geq$ 3) retained after the screening process described in Section 2 and summarized in Table 2, organized by focal construct: Narcissism (n = 88), Overconfidence (n = 86), and Hubris (n = 31). Five studies empirically test two of the three focal constructs within the same design and are therefore listed under both relevant constructs. References are ordered alphabetically by first author within each construct.

Narcissism (n = 88)

Agnihotri, A., & Bhattacharya, S. (2019). CEO Narcissism and Internationalization by Indian Firms. *Management International Review*.

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Agnihotri, A., & Bhattacharya, S. (2021). CEO narcissism and myopic management.

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Al-Shammari, M., Rasheed, A., & Al-Shammari, H. A. (2019). CEO narcissism and corporate social responsibility: Does CEO narcissism affect CSR focus? *Journal of Business Research*.

<https://doi.org/10.1016/j.jbusres.2019.07.005>

Al-Shammari, M., Rasheed, A. A., & Banerjee, S. N. (2022). Are All Narcissistic CEOs

Socially Responsible? An Empirical Investigation of an Inverted U-Shaped

Relationship Between CEO Narcissism and Corporate Social Responsibility. *Group &*

Organization Management. <https://doi.org/10.1177/10596011211040665>

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career horizon. *International Business Review*.

<https://doi.org/10.1016/j.ibusrev.2022.102078>

Bollaert, H., Leboeuf, G., & Schwienbacher, A. (2020). The narcissism of crowdfunding entrepreneurs. *Small Business Economics*.

<https://doi.org/10.1007/s11187-019-00145-w>

Brownell, K. M., Quinn, A., & Bolinger, M. T. (2024). The Triad Divided: A Curvilinear Mediation Model Linking Founder Machiavellianism, Narcissism, and Psychopathy to New Venture Performance. *Entrepreneurship Theory and Practice*.

<https://doi.org/10.1177/10422587231173684>

Brownell, K. M. (2026). Toxic or transformative? How lead founder machiavellianism, narcissism, and psychopathy shape new venture team survival. *Journal of Managerial Psychology*. <https://doi.org/10.1108/JMP-02-2025-0106>

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Appendix B

Number of Articles by Journal

Table B1 summarizes the distribution of the 200 unique articles included in the review across their 59 source journals, together with each journal's Academic Journal Guide (AJG) 2024 rating. The five studies that are dual-coded across two constructs (see Appendix A) are counted once here to avoid double-counting at the journal level.

Journal	AJG 2024 Rating	N Articles
Journal of Business Ethics	3	19
Journal of Business Research	3	15
Strategic Management Journal	4*	14
Journal of Management	4*	9
British Journal of Management	4	7
Journal of Business Venturing	4*	7
Leadership Quarterly	4	7
Small Business Economics	3	7

Academy of Management Review	4*	6
Business Strategy and the Environment	3	6
Journal of Management Studies	4	6
Management Science	4*	6
Administrative Science Quarterly	4*	5
International Business Review	3	5
Asia Pacific Journal of Management	3	4
Journal of Product Innovation Management	4	4
Journal of the Academy of Marketing Science	4*	4
Organization Science	4*	4
Research Policy	4*	4
European Management Review	3	3

IEEE Transactions on Engineering Management	3	3
Journal of Managerial Psychology	3	3
Journal of Small Business Management	3	3
Strategic Organization	3	3
California Management Review	3	2
Corporate Governance: An International Review	3	2
Entrepreneurship Theory and Practice	4*	2
European Journal of Marketing	3	2
Group & Organization Management	3	2
International Marketing Review	3	2
Internet Research	3	2
Journal of International Management	3	2
Journal of World Business	4	2

Management International Review	3	2
Technovation	3	2
Academy of Management Discoveries	3	1
Academy of Management Executive	4	1
Academy of Management Journal	4*	1
Academy of Management Perspectives	4	1
European Journal of Work and Organizational Psychology	3	1
Family Business Review	3	1
Harvard Business Review	3	1
Human Relations	4	1
Industrial Marketing Management	3	1
International Journal of Contemporary Hospitality Management	3	1

International Journal of Research in Marketing	4	1
International Small Business Journal: Researching Entrepreneurship	3	1
Journal of Accounting and Economics	4*	1
Journal of International Business Studies	4*	1
Journal of Marketing Research	4*	1
Journal of Technology Transfer	3	1
Management and Organization Review	3	1
Multinational Business Review	3	1
Organizational Dynamics	3	1
Personnel Psychology	4*	1
R & D Management	3	1
Strategic Entrepreneurship Journal	4	1

Technological Forecasting and Social Change	3	1
Tourism Management	4	1
Total	59 journals	200